

4(a)	$E = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 0.25 \times 2.3^2$	A1
	$= 0.66 \text{ J}$	
4(b)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
	$x = [(2 \times 0.66) / 420]^{0.5}$ $= 0.056 \text{ m}$	A1
	or	
	$E = \frac{1}{2}kx^2$	(C1)
	$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$ $x = (0.25 \times 2.3^2 / 420)^{0.5}$ $= 0.056 \text{ m}$	(A1)
4(c)(i)	$(p =) 0.25 \times 2.3$ or 0.25×1.5	C1
	change in momentum $= 0.25 (2.3 + 1.5)$ $= 0.95 \text{ N s}$	A1
4(c)(ii)	resultant force $= 0.95 / 0.086$ or $0.25 \times (2.3 + 1.5) / 0.086$ $= 11 \text{ N}$	A1
4(d)	curved line from the origin with an increasing gradient	B1